

# **Mathematics for Machine Learning**

MAI391

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# 1. Analytic Geometry

## 1.1 Norm

### Definition 1.1.1

A norm on  $V$  is a non-negative function

$$\|\cdot\| : V \rightarrow \mathbb{R} \quad (1.1)$$

such that  $\forall k \in \mathbb{R}$  and  $x, y \in V$ :

- *Absolute homogeneous*:  $\|kx\| = |k| \|x\|$
- **Triangle inequality**:  $\|x + y\| \leq \|x\| + \|y\|$
- **Positive definite**:  $\|x\| \geq 0$  and  $\|x\| = 0$  if and only if  $x = \vec{0}$



### Definition 1.1.2

Let  $p$  be a positive number. The  $l_p$ -norm  $\|\cdot\|_p$  in  $\mathbb{R}^n$ :

$$\|x\|_p = \sqrt[p]{\sum_{i=1}^n |x_i|^p} \quad \text{for any } x = \begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix} \in \mathbb{R}^n \quad (1.2)$$



### Example 1.1.3

- $p = 1 \rightarrow$  Manhattan norm  $l_1$ :

$$\|x\|_1 = \sum_{i=1}^n |x_i| \quad (1.3)$$

- $p = 2 \rightarrow$  Euclidean norm  $l_2$ :

$$\|x\|_2 = \sqrt{\sum_{i=1}^n |x_i|^2} \quad (1.4)$$

- $p = \infty \rightarrow$  maximum norm  $l_\infty$ :

$$\|x\|_\infty = \max\{|x_1|, \dots, |x_n|\} \quad (1.5)$$



**Definition 1.1.4**

Euclidean norm on  $\mathbb{R}^{m \times n}$ :

$$\|A\| = \left( \sum_{i=1}^m \sum_{j=1}^n a_{ij}^2 \right)^{\frac{1}{2}} \quad (1.6)$$

**Definition 1.1.5**

Spectral norm on  $\mathbb{R}^{m \times n}$ :

$$\|A\|_2 = \max_{x \in \mathbb{R}^n, \|x\|_2 = 1} \|Ax\|_2 \quad (1.7)$$



## 1.2 Inner Product

**Definition 1.2.1**

A linear mapping  $L$  on  $V$  is a function  $L : V \rightarrow \mathbb{R}$  that satisfies:

- $L(kx) = kL(x)$
- $L(x + y) = L(x) + L(y)$

$\forall x \in V$  and  $k \in \mathbb{R}$

**Definition 1.2.2**

A bilinear mapping or bilinear form  $B$  on  $V$  is a function  $B : V \times V \rightarrow \mathbb{R}$  that is **linear in each argument separately**:

- $B(kx, z) = kB(x, z)$
- $B(x + y, z) = B(x, z) + B(y, z)$
- $B(x, kz) = kB(x, z)$
- $B(x, y + z) = B(x, y) + B(x, z)$

$\forall x, y, z \in V$  and  $k \in \mathbb{R}$



**Problem 1.2.3**

Show that the mapping  $B : R^2 \times R^2 \rightarrow R$

$$B(x, y) = x_1y_1 - 2x_1y_2 + 3x_2y_2 \quad (1.8)$$

is a bilinear mapping on  $R^2$ .

◇

*Solution.*

$$\begin{aligned} B(kx, z) &= kx_1z_1 - 2kx_1z_2 + 3kx_2z_2 = k(x_1z_1 - 2x_1z_2 + 3x_2z_2) = kB(x, z) \\ B(x + y, z) &= (x_1 + y_1)z_1 - 2(x_1 + y_1)z_2 + 3(x_2 + y_2)z_2 \\ &= (x_1z_1 - 2x_1z_2 + 3x_2z_2) + (y_1z_1 - 2y_1z_2 + 3y_2z_2) \\ &= B(x, z) + B(y, z) \\ B(x, kz) &= kx_1z_1 - 2kx_1z_2 + 3kx_2z_2 = k(x_1z_1 - 2x_1z_2 + 3x_2z_2) = kB(x, z) \\ B(x, y + z) &= x_1(y_1 + z_1) - 2x_1(y_2 + z_2) + 3x_2(y_2 + z_2) \\ &= (x_1y_1 - 2x_1y_2 + 3x_2y_2) + (x_1z_1 - 2x_1z_2 + 3x_2z_2) \\ &= B(x, y) + B(x, z) \end{aligned}$$

$\Rightarrow B$  is linear mapping

□

**Definition 1.2.4**

A bilinear mapping  $B : V \times V \rightarrow R$  is called

- **symmetric** if  $B(x, y) = B(y, x)$
- **positive definite** if  $B(x, x) > 0$  for any  $V \ni x \neq \vec{0}$

♣

**Definition 1.2.5**

A positive definite, symmetric bilinear mapping  $B : V \times V \rightarrow R$  is called an **inner product**. We typically write  $\langle x, y \rangle$  instead of  $B(x, y)$  for inner product

♣

**Definition 1.2.6**

The pair  $(V, \langle \cdot, \cdot \rangle)$  is called **inner product space**

♣

**Definition 1.2.7**

The **dot product** or **scalar product** on  $\mathbb{R}^n$  is given by

$$x^T y = \sum_{i=1}^n x_i y_i \quad (1.10)$$

**Property 1.2.8**

Dot product is inner product, and  $(\mathbb{R}^n, \langle \cdot, \cdot \rangle)$  is **Euclidean vector space**

**Problem 1.2.9**

Consider  $V = \mathbb{R}^2$ . Define

$$\langle x, y \rangle = 2x_1 y_1 + x_1 y_2 + x_2 y_1 + 2x_2 y_2 \quad (1.11)$$

Show that  $\langle \cdot, \cdot \rangle$  is an inner product in  $\mathbb{R}^2$



*Solution.*

- $\langle \cdot, \cdot \rangle$  is a bilinear mapping (*The solution is similar to Problem 1.2.3*)
- $\langle \cdot, \cdot \rangle$  is symmetric

$$\begin{aligned} \langle x, y \rangle &= 2x_1 y_1 + x_1 y_2 + x_2 y_1 + 2x_2 y_2 \\ &= 2y_1 x_1 + y_1 x_2 + y_2 x_1 + 2x_2 y_2 = B(y, x) \end{aligned} \quad (1.12)$$

- $\langle \cdot, \cdot \rangle$  is positive definite

$$\begin{aligned} \langle x, x \rangle &= 2x_1^2 + 2x_2^2 + 2x_1 x_2 \\ &= (x_1 + x_2)^2 + x_1^2 + x_2^2 \geq 0 \end{aligned} \quad (1.13)$$

The equality occur only when  $x_1 = 0, x_2 = 0, x_1 + x_2 = 0$  which is  $x = \vec{0}$   
 $\Rightarrow \langle \cdot, \cdot \rangle > 0$  for any  $\mathbb{R}^2 \ni x \neq \vec{0}$ , or  $\langle \cdot, \cdot \rangle$  is positive definite

So  $\langle \cdot, \cdot \rangle$  satisfies all properties to be an inner product





**Definition 1.2.10**

A symmetric matrix  $A \in \mathbb{R}^{n \times n}$  is

- **positive definite** if

$$x^T A x > 0, \text{ for all non-zero vectors } x \in \mathbb{R}^n \quad (1.14)$$

- **positive semidefinite** if

$$x^T A x \geq 0, \text{ for all non-zero vectors } x \in \mathbb{R}^n \quad (1.15)$$

**Problem 1.2.11**

Consider the matrices

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \quad (1.16)$$

Show that  $A$  is positive definite but  $B$  is not



*Solution.*

$$\begin{aligned} x^T A x &= [x_1 \ x_2] \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= 2x_1^2 + 2x_1x_2 + 3x_2^2 \\ &> 0 \text{ for all non-zero vectors } x \in \mathbb{R}^n \end{aligned} \quad (1.17)$$

(the proof for why  $> 0$  is similar to [Problem 1.2.9](#))

Also  $A$  is symmetric so  $A$  is positive definite

$$\begin{aligned} x^T B x &= [x_1 \ x_2] \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= x_1^2 + 4x_1x_2 + 3x_2^2 \end{aligned} \quad (1.18)$$

Take  $x = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \Rightarrow x^T B x = -1$

So  $B$  is not positive definite and not positive semidefinite



**Note 1.2.12**

When multiply matrices of form  $x^T Ax$  (like in [Problem 1.2.11](#)), the coefficient of

$$x_i x_j = \begin{cases} a_{ij}, & \text{if } i = j \\ a_{ij} + a_{ji}, & \text{if } i \neq j \end{cases} \quad (1.19)$$



### 1.3 Lengths and Distances

**Definition 1.3.1**

The **norm** is  $\|x\| := \sqrt{\langle x, x \rangle}$

**Definition 1.3.2**

The **distance between  $x$  and  $y$**  is  $d(x, y) := \|x - y\| = \sqrt{\langle x - y, x - y \rangle}$

**Theorem 1.3.3**

**Cauchy-Schwarz inequality**

$$|\langle x, y \rangle| \leq \|x\| \|y\| \quad (1.20)$$

**Problem 1.3.4**

Let the inner product  $\langle \cdot, \cdot \rangle$  be defined on  $\mathbb{R}^n$  by

$$\langle x, y \rangle = x_1 y_1 - x_1 y_2 - x_2 y_1 + 3x_2 y_2 \quad (1.21)$$

Find the distance between  $u = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$  and  $v = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$



*Solution.*

$$u - v = \begin{bmatrix} 2 \\ -1 \end{bmatrix} \quad (1.22)$$

$$\begin{aligned}
d(u, v) &= \|u - v\| = \sqrt{\langle u - v, u - v \rangle} \\
&= \sqrt{2 \cdot 2 - 2 \cdot (-1) - (-1) \cdot 2 + 3 \cdot (-1) \cdot (-1)} \\
&= \sqrt{11}
\end{aligned} \tag{1.23}$$

□

## 1.4 Angles and Orthogonality

### Definition 1.4.1

The **angle** between vector  $x$  and  $y$  is the number  $\theta \in [0, \pi]$  defined by

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|} \tag{1.24}$$

♣

### Problem 1.4.2

Define the inner product  $\langle \cdot, \cdot \rangle$

$$\langle x, y \rangle = x \cdot y = x_1 y_1 + x_2 y_2 \text{ (dot product)} \tag{1.25}$$

Find the angle between  $x = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$  and  $y = \begin{bmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}$

◇

*Solution.*

$$\langle x, y \rangle = 1 \cdot \left(-\frac{1}{2}\right) + 0 \cdot \frac{\sqrt{3}}{2} = -\frac{1}{2} \tag{1.26}$$

$$\begin{aligned}
\|x\| &= \sqrt{1 \cdot 1 + 0 \cdot 0} = 1 \\
\|y\| &= \sqrt{-\frac{1}{2} \cdot -\frac{1}{2} + \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2}} = 1
\end{aligned} \tag{1.27}$$

$$\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|} = \frac{-\frac{1}{2}}{1 \cdot 1} = -\frac{1}{2} \tag{1.28}$$

$$\Rightarrow \theta = \cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3} \tag{1.29}$$

□

**Definition 1.4.3**

Two vectors  $x$  and  $y$  are **orthogonal** (or  $x \perp y$ )  $\Leftrightarrow \langle x, y \rangle = 0$

**Definition 1.4.4**

If  $x \perp y$  and  $\|x\| = \|y\| = 1$  ( $x$  and  $y$  are unit vectors) then  $x$  and  $y$  are **orthonormal**

**Definition 1.4.5**

A square matrix  $A \in \mathbb{R}^{n \times n}$  is an **orthogonal matrix** if and only if

$$AA^T = A^T A = I_n \quad (1.30)$$

**Property 1.4.6**

If  $A \in \mathbb{R}^{n \times n}$  is orthogonal then

- $A^{-1} = A^T$
- $A$  preserve the length of any vector  $x \in \mathbb{R}^n$

$$\|Ax\| = \|x\| \quad (1.31)$$

- $A$  preserve the angle between two vectors  $x$  and  $y$  in  $\mathbb{R}^n$

$$\text{angle}(Ax, Ay) = \text{angle}(x, y) \quad (1.32)$$

**Note 1.4.7**

For square matrix  $A \in \mathbb{R}^{n \times n}$ ,  $A^{-1}$  is defined as matrix with

$$AA^{-1} = A^{-1}A = I_n \quad (1.33)$$

So if  $A$  is orthogonal,  $A^T = A^{-1}$



## 1.5 Orthonormal Basis

### Definition 1.5.1

Given a vector space  $V$ , a set of vectors  $\{v_1, \dots, v_n\}$  is **linear independent** if the **only solution** to equation

$$a_1 v_1 + \dots + a_n v_n = 0 \quad (1.34)$$

is  $a_1 = \dots = a_n = 0$



### Definition 1.5.2

Given a vector space  $V$ , a set of vectors  $B = \{b_1, \dots, b_n\}$  is basis if:

- **Every subset** of  $B$  is **linear independent**
- For every  $x \in V$  there exist  $\{a_1, \dots, a_n\}$  such that

$$a_1 b_1 + \dots + a_n b_n = x \quad (1.35)$$



### Example 1.5.3

Most common basis for  $\mathbb{R}^2$  is  $B = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$



### Definition 1.5.4

Consider an dimensional inner product vector space  $(V, \langle \cdot, \cdot \rangle)$  and a basis  $B = \{b_1, \dots, b_n\}$  of  $V$ .  $B$  is **orthonormal basis** if

$$\langle b_i, b_j \rangle = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (1.36)$$



### Theorem 1.5.5

Let  $B = \{b_1, \dots, b_n\}$  be an orthonormal basis of an inner product space  $V$ . If  $x$  is any vector in  $V$ , then

$$x = \langle x, b_1 \rangle b_1 + \dots + \langle x, b_n \rangle b_n \quad (1.37)$$



**Problem 1.5.6**

Define inner product

